

PAPER_858: Westerlund 2 Quadriadic UQFF Four-Set Simultaneous Real and Imaginary Solutions

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-04 **Session:** 199 **Source:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 - Aug 07, 2025) **Calculator:** Westerlund2QuadriadicRealImaginaryCalc (CP4 #442) **CVW:** v2.0.0 compliant

Abstract

We solve the full quadriadic UQFF system (four master equations) simultaneously for Westerlund 2, a massive star cluster in Carina ($\sim 20,000$ ly, age ~ 2 Myr). The four equations - (1) Compressed Gravity F_{U_g} , (2) Resonance $R(t)$, (3) Buoyancy $F_{U_{Bi}}$, (4) Universal Magnetism U_m - each yield both real and imaginary solutions. The imaginary components represent the quantum superconducting portion of each field. For Westerlund 2: $F_{U_g} \sim 2.43e-40$ N, $R(t) \sim 6.67e-41$ N, $F_{U_{Bi}} \sim 6.14e-32$ N, $U_m \sim 1.80e5$ J/m³. Buoyancy dominates the force balance, consistent with the region's intense star formation activity.

1. Core Equations

- $F_{U_g} = \sum_k [k_k (f_{UA}' f_{SCm} REB)^{2/r^2} * G_k] + k_4 \rho_{vac} M_{BH}/r * \exp(-\alpha * t) * \cos(\pi * t/n) * (1 + f_{feedback})$
 - $R(t) = \sum_{i=1}^{26} [R_{Ug1,i} \cos(w_1 * t) + R_{Ug2,i} \cos(w_2 * t) + R_{Ug3,i} \cos(w_3 * t) + R_{Ug4,i} \cos(w_4 * t)]$
 - $F_{U_{Bi}} = \sum_k [k_{Ub,k} f_{UA}' f_{SCm} REB / r^2 * H_k * f_{Ub}]$
 - $U_m = \sum_j [\mu_j(t) / r_j * (1 - \exp(-\gamma * t) * \cos(\pi * t/n)) * \phi^j] * P_{SCm} * E_{react} * (1 + 1e13 * f_H) * (1 + f_q)$
 - $f_{Ub} = k_{Ub} * \Delta_{k_{eta}} * (\rho_{vac}, [UA] / \rho_{vac}, [SCm]) * (V_{little} / V_{big})$
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2. UQFF Integration

This calculator operates as a stateless physics calculator within the Condensed-Physics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py -> bodies_*.csv data flow.

3. Source Data

- File:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 - Aug 07, 2025)
 - Session:** 199
 - VDS/DVP/BH:** ABSENT (general vacuum density references only)
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$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}} \text{ suppression}$	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Srivastava, Y.N., Widom, A., Larsen, L. – Electroweak neutron production (LENR)
3. Kepler Mission DR25 – 4,034 candidates, 2,335 confirmed planets
4. Hubble Heritage Team / A. Nota (ESA/STScI) – Westerlund 2 / NGC 346 imaging
5. UQFF Calibration: $\kappa=0.0005/\text{day}$, $[\text{SSq}]=0.57$, $\beta_i \sim 0.603$